1. Identity

Document	STAR-PKG-CONJUNCTION-SCREEN-SGP4-EN · v2.0 · 2026-10-21
Package	conjunction-screen-sgp4
Version	0.2.1
Capability	star.conjunction.screen
Maturity	VERIFIED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/conjunction_screen_sgp4

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

NEEDS	LICENCE	SOURCE
numpy	BSD-3-Clause	https://numpy.org/
scipy	BSD-3-Clause	https://scipy.org/
sgp4	MIT	https://pypi.org/project/sgp4/

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())"
conjunction_screen_sgp4-0.2.1-py3-none-any.whl
```

FILE	SHA-256
conjunction_screen_sgp4-0.2.1-py3-none-any.whl	04d5e0965645bdac104be9202995eaf6fefe640651b1da8f853499c0ff4e2d65
conjunction_screen_sgp4-0.2.1-source.zip	883bbc4727726bf6030c0e43980c24af2a74537d888ffc7329b246fc5a4545ab

5. What it does, and its limits

Screening of close approaches between objects of TLE catalogues (who to look at), NOT a collision-probability tool.

Probability of collision, covariance, manoeuvre advice, accuracy below the ~1 km TLE error. Advisory screening only.

- `screen(tle_paths, start, hours, step, coarse_km, final_km, cross_only, refine)` returns encounters (TCA in UTC, miss distance km, age of the older TLE) below `final_km`, excluding pairs that stay close (formations).
- TLEs are propagated with SGP4 only (mean elements); propagation errors are reported, never replaced.
- `refine_tca(state_at, step)`: closed-form TCA from relative motion, shared by screening and tests.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_conjunction_screen_sgp4_0_2_1,
  author  = {Cavazzini, Andrea},
```

```
  title   = {{conjunction-screen-sgp4}: Conjunction screening of TLE catalogues with SGP4, KD-
tree coarse pass and closed-form TCA refinement},
  version = {0.2.1},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/conjunction_screen_sgp4/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts